SCPN-Fusion-Core: A Hybrid Rust–Python Grad–Shafranov Solver with Evidence-Bounded Reconstruction and Neural Fallback

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Abstract

We describe the equilibrium-solver architecture of SCPN-Fusion-Core, an open-source Python and Rust toolkit for axisymmetric tokamak modelling. The implementation contains fixed- and free-boundary Grad–Shafranov paths, Picard iteration, successive over-relaxation and multigrid components, synthetic inverse reconstruction, and an optional neural-surrogate path with explicit backend and fallback reporting. This paper deliberately separates implemented capability from demonstrated accuracy. The packaged pointwise benchmark records 16/16 acceptance-gated SPARC rows on the neural backend under the declared pointwise error threshold. Only 8/16 also pass the adapted source-convention contract, while diagnostic non-SPARC proxy rows remain outside that acceptance cohort. This is not a free-boundary or experimental-accuracy result. Separately, a provenance-bound *same-case*, warm-start reproduction of DIII-D EFIT shot 145419 at 2100 ms has 1.908% full-domain deep-region RMS flux error after 21 Anderson-accelerated iterations. The corresponding plasma-free cold start remains at 126.788% deep RMS and cannot select the physical branch. These are reproduction and failure-characterisation results, not blind equilibrium prediction. Earlier quantitative accuracy and portable speedup claims are not retained because the current evidence does not support them under a common full-fidelity contract. We report the useful software, the negative results, and the validation work still required.

Keywords: Grad–Shafranov equation; tokamak equilibrium; inverse reconstruction; Rust; Python; fail-closed surrogate; reproducibility

1 Scope and claim boundary

Software version and date. This preprint describes SCPN-Fusion-Core at evidence revision `476908debdd8` of 26 August 2026, on the development line between releases 4.0.0 and 4.1.0. On 13 November 2026 the DIII-D reproduction and the SPARC pointwise benchmark were rerun on the release 4.1.0 line (commit `36326e5eec27`); every quantity reported below was reproduced at its printed precision, and the SPARC rows reproduced exactly. Corrections made since the review draft are listed in the dated note at the end of the paper.

The Grad–Shafranov (GS) equation is the standard axisymmetric magnetohydrostatic equilibrium equation for a toroidal plasma [Grad and Rubin, 1958, Shafranov, 1966]. Equilibrium reconstruction is mature and is served by established codes such as EFIT [Lao et al., 1985, 1990] and FreeGS [Dudson, 2019]. A new implementation therefore needs more than a broad feature list: it needs

same-case numerical evidence, transparent conventions, failure semantics, and comparisons that hold input physics and tolerances fixed.

This paper is a software and evidence report bound to evidence revision `476908debdd8`. The full revision and the later package-preparation base are recorded in the paper metadata. It does not claim that SCPN-Fusion-Core replaces EFIT, FreeGS, or a production plasma-control system. It does not claim experimental predictive accuracy, a portable performance lead, or production free-boundary neural-surrogate admission. Synthetic figures are labelled as illustrations; real-data quantities are tied to the packaged JSON evidence.

2 Mathematical formulation

In cylindrical coordinates (R, ϕ, Z), axisymmetric equilibrium satisfies

$$\Delta^* \Psi = -\mu_0 R^2 p'(\Psi) - F(\Psi) F'(\Psi), \tag{1}$$

where Ψ is poloidal flux, p is pressure and $F = RB_\phi$. The elliptic operator is

$$\Delta^* \Psi = R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \Psi}{\partial R} \right) + \frac{\partial^2 \Psi}{\partial Z^2}. \tag{2}$$

The source term depends nonlinearly on the flux through p' and FF'. The implementation therefore performs an outer fixed-point iteration: evaluate the source from the current flux, solve the linear elliptic problem, and relax the update. Fixed-boundary and free-boundary problems share numerical components but do not have identical physical inputs. In particular, a free-boundary reconstruction requires external-coil currents or an equivalent measured external-source description; omitting them changes the problem.

3 Implementation

3.1 Hybrid architecture

The Python layer owns configuration, reference-data parsing, experiment orchestration and high-level analysis. Numerical kernels are also available in a Rust workspace and are exposed to Python through PyO3 [PyO3 Contributors, 2024]. The repository configures a hosted matrix for Python 3.10–3.12, Rust, the inter-language boundary, security checks and scientific gates. This paper does not claim exact-head hosted success from that configuration alone. Build and interface checks would not by themselves establish physical validity.

3.2 Forward solution

The discretisation represents Eq. (1) on an (R, Z) mesh. The code contains red–black successive over-relaxation and multigrid components inside a Picard outer loop. Boundary handling, sign conventions and normalisation are treated as explicit contracts because a low residual under the wrong source convention is not a valid equilibrium result.

3.3 Inverse reconstruction

The inverse path adjusts a finite parameter vector to reduce residuals between modelled and supplied diagnostic values. It uses Levenberg–Marquardt updates [Levenberg, 1944, Marquardt,

1963] with Tikhonov regularisation [Tikhonov and Arsenin, 1977] of the parameter step. Figure 1 is a deterministic synthetic probe example. It illustrates the spatial input and residual layout; it does not execute the inverse solver or recover a real diagnostic system.

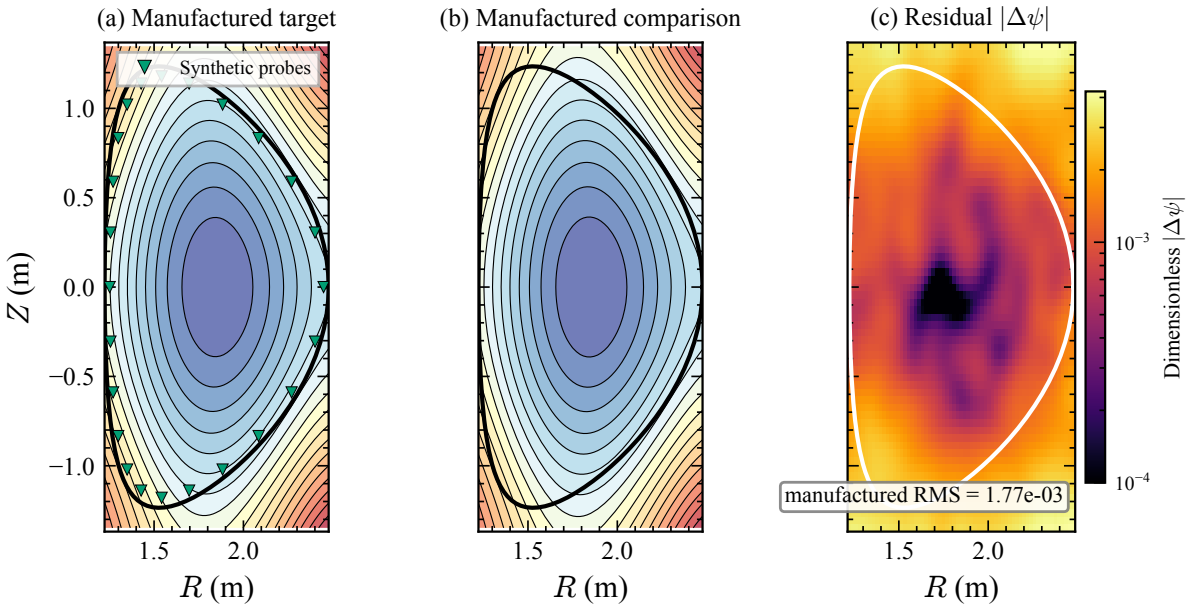


Figure 1: Deterministic manufactured inverse-reconstruction illustration. Target flux, perturbed comparison flux and residual are generated from synthetic probe locations. This figure is not experimental validation.

3.4 Neural path and failure semantics

The repository contains a neural-equilibrium interface with explicit backend reporting. The packaged pointwise artifact records 16 acceptance-gated SPARC rows, all executed on the neural backend and all passing the declared 5% NRMSE threshold. Eight of those 16 rows also pass the adapted source-convention contract. Its diagnostic non-SPARC proxy rows are outside that gated cohort and may use a reduced-order fallback. This establishes the bounded pointwise software gate only: it does not establish training-data origin, independent free-boundary prediction, experimental accuracy, uncertainty calibration or latency. Those claims require immutable training provenance, device- or shot-disjoint calibration and test partitions, calibrated uncertainty and a same-case comparison against a production solver.

4 Evidence

4.1 Illustrative SPARC-like geometry

Figure 2 uses published SPARC-like geometry parameters [Creely et al., 2020] in a dimensionless manufactured flux field. It is useful for visualising a shaped geometry, but it is not a Grad–Shafranov solution, a solved SPARC discharge or part of the quantitative validation table.

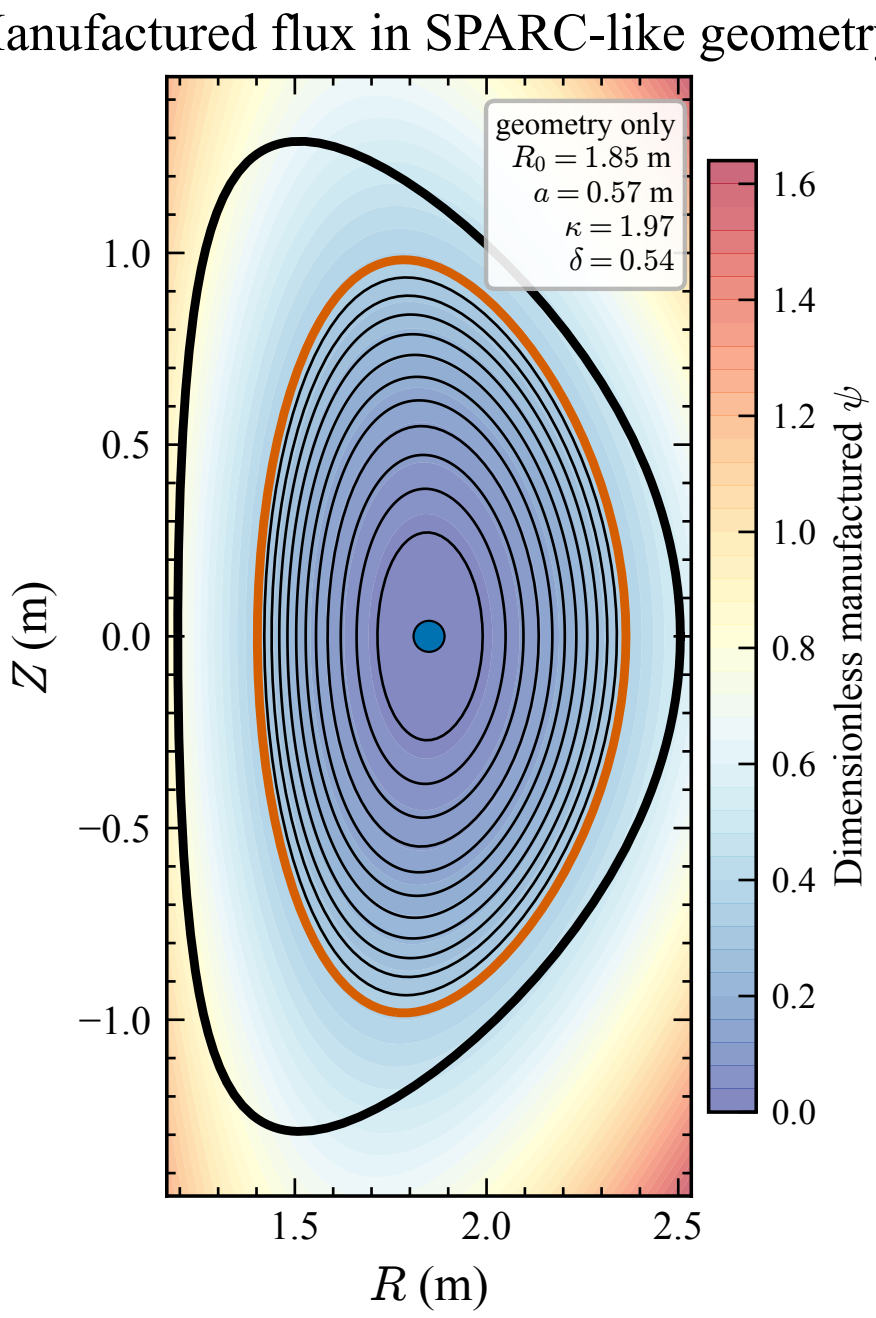


Figure 2: Dimensionless manufactured flux in SPARC-like geometry. The geometry is illustrative; the contours are not a Grad–Shafranov solution, experimental reconstruction or predictive reconstruction.

4.2 Real DIII-D same-case reproduction

The packaged evidence file `evidence/real_diiid_145419_validation.json` identifies the reference as the public OMAS sample `g145419.02100`, DIII-D shot 145419 at 2100 ms, and records its SHA-256 hash, generator hash, dependency versions and modelling disclosure. The reproduction lanes initialise from measured EFIT flux and use reference-derived normalisation anchors. They are consistency tests, not blind prediction or independent branch selection.

Table 1: Selected exact quantities from the packaged DIII-D evidence. Errors are normalised by the reference poloidal-flux span.

Lane	Deep RMS	Global maximum	Iterations
Warm-start, full-domain reproduction	1.908%	3.908%	21
Warm-start, 4×4 sub-cell source	1.833%	3.729%	20
Plasma-free cold start	126.788%	173.139%	1
Shell-pinned attribution diagnostic	0.0697%	0.1741%	14

The shell-pinned row uses additional measured information and is included only to localise the remaining mismatch; it is not a model improvement. The cold start is the decisive negative result: fixed reference anchors leave the plasma-free state stationary, so the reproduction map cannot discover the physical branch. Predictive free-boundary work remains blocked on complete external-coil input and independent branch-selection evidence.

4.3 SPARC and broader validation boundary

The packaged pointwise artifact records its neural backend, acceptance threshold and source-convention blockers. All 16 gated rows pass the pointwise NRMSE and backend requirements, while only 8 of 16 pass the adapted source-convention contract. The GEQDSK contract does not supply the same-case external-coil current description needed for a predictive free-boundary reconstruction. Consequently the result is not promoted to a free-boundary or experimental-accuracy claim. Likewise, confinement-scaling and disruption datasets exercise different subsystems and do not validate the equilibrium reconstruction simply because they share the same repository.

5 Performance and comparisons

No current benchmark applies one full-fidelity numerical contract to the Python, Rust and external-solver lanes. This paper therefore makes no hardware-neutral speed claim. A publishable comparison must use the same equilibrium, source convention, boundary data, precision, convergence tolerance and failure accounting; report both cold and warm distributions; identify the CPU/GPU, compiler and thread settings; and retain divergent cases. The same requirements apply to any comparison with FreeGS, EFIT or another community code.

6 Limitations and required validation

The current evidence leaves several important gaps:

- the real-data result is warm-started same-case reproduction rather than blind reconstruction;
- one public DIII-D shot/time slice is not multi-machine validation;

- external-coil currents are required for predictive free-boundary branch selection and are not inferred from geometry alone;
- SPARC source-convention and missing external-coil blockers remain visible;
- the pointwise neural gate is not production or free-boundary admission;
- no fair, current, same-case performance comparison is included;
- no hardware-in-the-loop or production PCS claim is made.

7 Reproducibility

The manuscript source, references, figure generators and evidence JSON are in this directory. `BUILD.md` gives the exact LaTeX and figure commands. The manifest binds the DIII-D payload to a tracked source at evidence revision `476908debdd8`, including its Git object and exact file hash. The SPARC source artifact was Git-ignored when generated; its exact source hash, generation time and generator hash are retained, and the paper-local JSON is therefore the authoritative public custody copy. The final PDF is an inspected artefact generated from this source. The manuscript-content licence has not yet been selected by the author; repository code retains its existing licence.

Note added (13 November 2026)

This note lists every change from the review draft of 4 September 2026.

- The review draft stated that the inverse path includes a Huber loss. That was wrong for the bound revision: a Huber loss existed in the repository only on 12 February 2026 and was replaced the same day by the Tikhonov-regularised Levenberg–Marquardt step described above. The sentence and its citation are corrected.
- The hosted matrix named in the implementation section is the one at the bound revision. Release 4.1.0 removed Python 3.10; the matrix now covers Python 3.11 and 3.12.
- The author affiliation now reads ANULUM CH&LI, Marbach SG, Switzerland.
- The figures were regenerated with the plotting library pinned at release 4.1.0; only the typesetting of axis tick labels changed.
- Sentences that called this document a review draft now call it a paper or preprint.

8 Conclusion

SCPN-Fusion-Core provides a substantial hybrid implementation of axisymmetric equilibrium workflows, but implementation breadth is not a substitute for validation. The packaged DIII-D same-case result demonstrates a useful warm-start reproduction and, equally importantly, a plasma-free cold-start failure. The gated SPARC pointwise result and diagnostic fallback behavior are reported without promoting either to free-boundary accuracy. These boundaries define the next work: complete physical inputs, independent branch selection, multi-machine held-out evidence and fair same-case comparisons.

Data and code availability

Code is available at https://github.com/anulum/scpn-fusion-core. This preprint is bound to evidence revision `476908debdd8`; the full revision is recorded in the paper metadata. Paper-local evidence and reproduction instructions are distributed beside the manuscript.

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